Discussion 3: Face Recognition

Discussion Topic:

The Good, the Bad, & the Ugly Face Challenge Problem encourages the development of algorithms that are robust to recognition across changes that occur in still frontal faces. This challenge has three partitions. The first, a Good partition, contains pairs of images that are easily recognizable. Next, we have the Bad. This partition contains image pairs of average difficulty to recognize. Finally, the images that are considered difficult to recognize are the Ugly partition.

Take a look at the following figures:

Figure 1

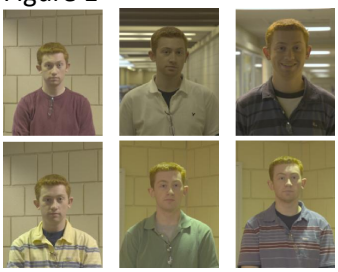


Figure 2

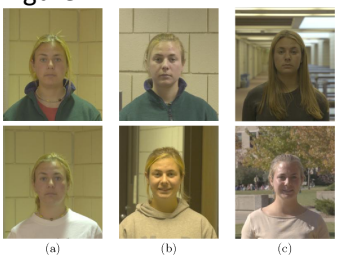


Figure 3



Each pair has been partitioned. For one of the figures, address the following:

Discuss which partition (good, bad, ugly) you think each pair (a, b, c) belongs to. Substantiate your claim with at least two reasons you are placing the image pair in each partition.
What image processing techniques discussed so far could be performed to make Bad and Ugly partitions easier for a recognition system to recognize?

My Post:

Hello Class

Hello Class,

I would place pair (a) in Good, pair (b) in Bad, and pair (c) in Ugly.
I assess the faces based on how I, a human, perceive human faces, and on my understanding of how similarity algorithms work; see Table 1. However, human perception and algorithmic similarity are not always the same. What looks easy to us is not necessarily easy for a recognition system (Phillips et al., 2012).

I have to say, the wording of the assignment topic was a bit confusing to me. At first glance, I thought it was about recognizing facial expressions. After reading it a couple of times, I realized it was about comparing the images' paired faces (a, b, c) and assessing how difficult it would be for a facial recognition system to compare paired human faces…

Table 1
My Observations

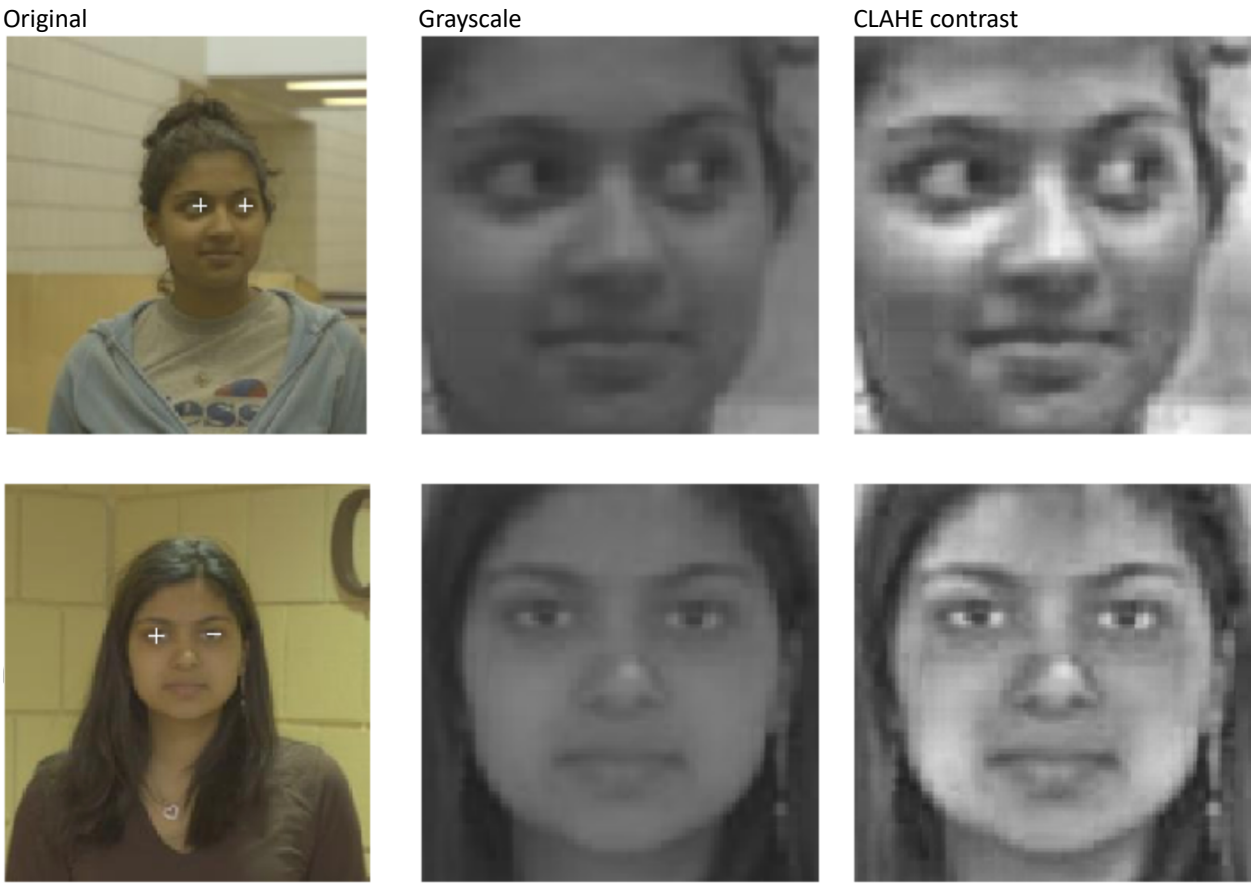
Figure	Pair	Partition	My Observations
1	**(a)**	**Good**	To me, these two images look fairly easy to compare. Both faces are close to frontal, the expressions are mostly neutral, and the faces are about the same size. The eyes, nose, and mouth are also clearly visible in both images.
1	**(b)**	**Bad**	I see more differences in this pair. The top image has a more neutral expression, while the bottom image shows a smile, which changes the mouth and cheeks. The bottom image is also brighter and has warmer lighting.
1	**(c)**	**Ugly**	This pair looks much harder to compare. The top image is indoors and the face is partly in shadow, while the bottom image is outdoors with much stronger lighting. The expression also changes from neutral to a smile, so correcting the brightness alone would not make the two faces look the same.
2	**(a)**	**Good**	These two images look fairly similar to me. Both faces are close to frontal, the expressions are mostly neutral, and the face sizes are similar. The clothing is different, but I would crop most of that out and focus on the face.
2	**(b)**	**Bad**	The first thing I notice is that the top face is darker than the bottom one. The face is also a different size and position in the two images. Because of that, I think the faces would need to be aligned before comparing the same facial areas directly.
2	**(c)**	**Ugly**	The biggest difference I see is the expression. The top image has a broad smile, while the bottom image is much more neutral. The lighting is also different, especially around the forehead and nose.
3	**(a)**	**Good**	Both faces are still close to frontal, and the main facial features are easy to see. The face sizes are also fairly similar. There are some differences in lighting and expression, but overall the two faces still look reasonably consistent to me.
3	**(b)**	**Bad**	The top image shows a smile, while the bottom image is more neutral. I also notice that the face is not exactly the same size or in the same position in the frame. Even though the backgrounds look similar, the facial regions are still different.

3	**(c)**	**Ugly**	This pair looks the most difficult to me in Figure 3. In the top image, the person is looking more to the side and the head is slightly turned, while the bottom image is more frontal. The hair also changes the outline around the face, so the two images do not line up as well visually.

To pre-process the image using OpenCV. First, I would clip the faces from the images and align them with `cv2.warpAffine()`. Note that the eyes are used as reference points. In the example below, I used Figure 3 Pair (c); I marked the eyes manually, and based on the eye position, I cropped each image to 128 × 128 pixels. Then, `cv2.cvtColor()` converts the crops to grayscale (OpenCV, n.d.-a, n.d.-b). Additionally, to address lighting, which is different between the two images and needs to be addressed using a normalization method. I would use `cv2.equalizeHist()` or Contrast-Limited Adaptive Histogram Equalization (CLAHE), `cv2.createCLAHE()`. In the example, I used CLAHE, which adjusts contrast in smaller regions and handles excessive contrast amplification (OpenCV, n.d.-b).

Figure 1

Example Figure 3 Pair (c)



I would compare the CACH images using `cv2.addWeighted()` to overlap/blend the images for display purposes, and `cv2.absdiff()` to compare pixel-by-pixel the absolute difference, which is the difference in grayscale intensity (light intensity) between the two images; see Figure 2 and the code snippet. Note that these functions are used for pre-processing the image; they do not produce a facial-recognition result or similarity score.

Figure 2
Pre-process Overlay



Code Snippet 1

```
# The first image's pixel is multiplied by the alpha weight = 0.5.
# The second image's pixel is multiplied by the beta weight = 0.5.
# Gamma = 0 adds zero to the combined alpha and beta values, no additional
# brightness is added.
overlay = cv2.addWeighted(top_figure3c_cach, 0.5, bottom_ top_figure3_cach,
0.5, 0)

# absdiff means absolute difference
# difference[y,x]=|top_processed[y,x]-bottom_processed[y,x]|
# This will show how much the images' intensities differ
difference = cv2.absdiff(top_processed, bottom_processed)
```

The overlay lets me inspect how well the images' corresponding regions line up. In the "Intensity difference" image, black means equal pixel values, while brighter pixels indicate larger intensity differences. In the example, we can see that the differences remain around the eyes and mouth.

The actual recognition step would use the pre-processed images. From them, a recognition algorithm would extract facial features and compare their numerical representations to calculate a similarity score. preprocessing would also need to suit the selected model. This demonstration stops at preprocessing and visual inspection; it does not establish a successful facial match.

References

Phillips, P. J., Beveridge, J. R., Draper, B. A., Givens, G., O'Toole, A. J., Bolme, D., Dunlop, J., Lui, Y. M., Sahibzada, H., & Weimer, S. (2012). The good, the bad, and the ugly face challenge problem. *Image and Vision Computing, 30*(3), 177–185. https://doi.org/10.1016/j.imavis.2012.01.004

OpenCV. (n.d.-a). *Changing colorspaces*. https://docs.opencv.org/4.13.0/df/d9d/tutorial_py_colorspaces.html OpenCV Documentation

OpenCV. (n.d.-b). *Geometric transformations of images*. https://docs.opencv.org/4.13.0/da/d6e/tutorial_py_geometric_transformations.html OpenCV Documentation